QUALIFICATION REQUIREMENTS REMOVED

MIL-M-38510/403A
16 August 1983
SUPERSEDING
MIL-M-38510/403
14 June 1982

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, NMOS,
2048-WORD x 8-BIT STATIC READ ONLY MEMORY (ROM),
MONOLITHIC SILICON

INACTIVE FOR NEW DESIGN
AFTER DATE OF THIS REVISION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, silicon gate, NMOS, ROM microcircuits. One product assurance class and a choice of case outlines and lead finishes are provided and are reflected in the part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510 with the exception that the "JAN" or "J" certification mark shall not be used.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	2048-word x 8-bit static ROM with 3-state outputs

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
J	D-3 (24-lead, 1/2" x 1-1/4") dual-in-line package

1.3 Absolute maximum ratings.

Storage temperature range	-65°C to +150°C
Case operating temperature range (T_C)	-55°C to +125°C
Supply voltage range	-0.3 V dc to +7.0 V dc
Power dissipation (P_D) (for T_C = -55°C to +125°C)	1.0 W maximum
Input voltage range, all inputs	-0.3 V dc to +7.0 V dc
Lead temperature (during soldering, at distance 1/16 ±1/32 inch (1.59 ±0.79 mm) from case for 10 seconds maximum)	+265°C
Thermal resistance, junction-to-case (θ_{JC})	0.0833°C/mW
Maximum junction temperature (T_J)	+150°C 1/

1.4 Recommended operating conditions (at T_C = -55°C to +125°C).

Supply voltage range (V_{CC})	+4.5 V dc to +5.5 V dc
Low level input voltage range (V_{IL})	-0.3 V dc to +0.8 V dc
High level input voltage range (V_{IH})	+2.0 V dc to V_{CC}

1/ Maximum junction temperature (T_J) may be increased to 175°C during the burn-in and steady-state life tests.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5962

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2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein. An altered item drawing shall be prepared by the contracting activity to specify the required mask-program.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Block diagram. The block diagram shall be as specified on figure 2.

3.2.3 Case outline. The case outline shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 User mask-program. Since the ROM is programmed by the manufacturer in a wide variety of configurations, the contracting activity shall provide an altered item drawing describing the mask-program to be used by the manufacturer. All testing requirements and quality assurance provisions herein, including the requirements of the altered item drawing, shall be satisfied by the manufacturer prior to delivery.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table II and, where applicable, on the altered item drawing. The electrical tests for each subgroup are described in table III.

3.7 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. The altered item drawing number describing the user mask-program shall be added to the marking by the manufacturer. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container. The "JAN" or "J" certification mark shall not be used.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ T_C = -55°C to +125°C	Limits Min	Limits Max	Unit
Low-level output voltage	V_{OL}	I_{OL} = 1.6 mA	---	0.4	V
High-level output voltage	V_{OH}	I_{OH} = -100 µA	2.4	---	V
Input leakage current	I_{IL}	V_{IN} = 0 V	---	2.5	µA
	I_{IH}	V_{IN} = 5.5 V	---	2.5	
Output leakage current	I_{OLZ}	V_{OUT} = 0.4 V, chip deselected	---	10	µA
	I_{OHZ}	V_{OUT} = 2.4 V, chip deselected	---	10	
Power supply current	I_{CC}	V_{CC} = 5.5 V	---	130	mA
Read cycle time	TAVAX	See waveforms, table III, footnote 9/	450	---	ns
Address access time	TAVQV		---	450	ns
Enable access time	TEVQV		---	200	ns
Output disable time	TEXQZ		10	150	ns
Input capacitance	C_i	V_{IN} = 0 V	---	7.5	pF
Output capacitance	C_o	V_{IN} = 0 V	---	12.5	pF

1/ See table III for exact pin test conditions.

3.8 Manufacturer eligibility. To be eligible to supply microcircuits to this specification, a manufacturer shall have a manufacturer certification in accordance with MIL-M-38510 for at least one line; not necessarily the line producing the device type described herein.

3.9 Certification. Certification in accordance with MIL-M-38510 is not required for this device.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test Condition D or E, using the circuit shown on figure 3 or equivalent.

(2) T_A = +125°C minimum.

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b. Dynamic test (test conditions D and E) using the circuit shown on figure 4, or equivalent. The test duration shall be 240 hours minimum at +125°C. Interim electrical measurements shall be made between the static test and dynamic test, and shall consist of subgroup 1 of table III. Measurements shall be on a pass-fail basis only and shall not be used for delta limits.

For class B devices, dynamic burn-in shall be performed for 160 hours minimum at +125°C ambient temperature, using the circuit shown on figure 4, or equivalent.

c. The percent defective allowable (PDA) for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in, divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

d. Interim and final electrical test parameters shall be as specified in table II.

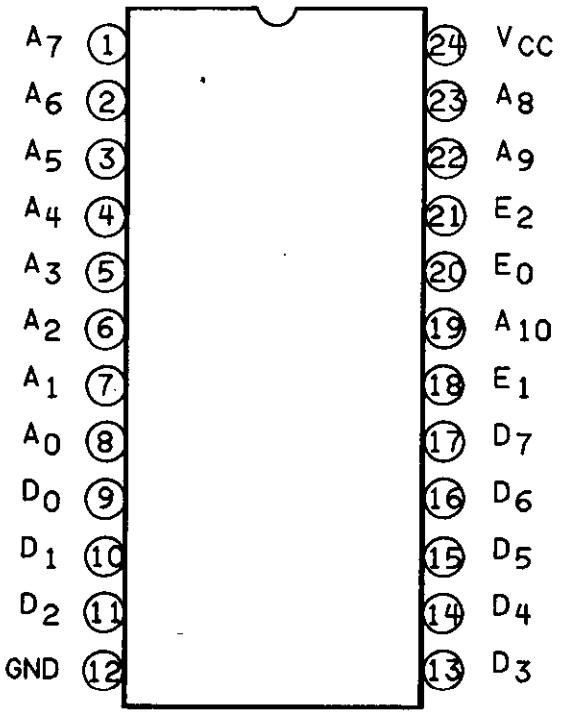
e. All devices shall be mask-programmed to the requirements of the altered item drawing prior to initiation of any testing.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)
	Class B devices
Interim electrical test parameters (pre burn-in) (method 5004)	1, 7
Interim electrical test parameters (between static burn-in and dynamic burn-in) (method 5004)	1*
Final electrical test parameters (method 5004)	1*, 2, 3, 7*, 8, 9, 10, 11
Group A test requirements (method 5005)	1, 2, 3, 7, 8, 9, 10, 11
Group B electrical tests (method 5005)	1, 2, 3, 7, 8, 9, 10, 11
Group C end-point electrical tests (method 5005)	NA
Additional subgroups for group C inspection	None
Group D end-point electrical tests (method 5005)	1, 2, 3, 7, 8, 9

* PDA applies to subgroups 1 and 7 (see 4.2c).

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A_0-A_{10}: Address Inputs
D_0-D_7: Data Outputs
E_0-E_2: Chip Select Inputs

FIGURE 1. Terminal connections.

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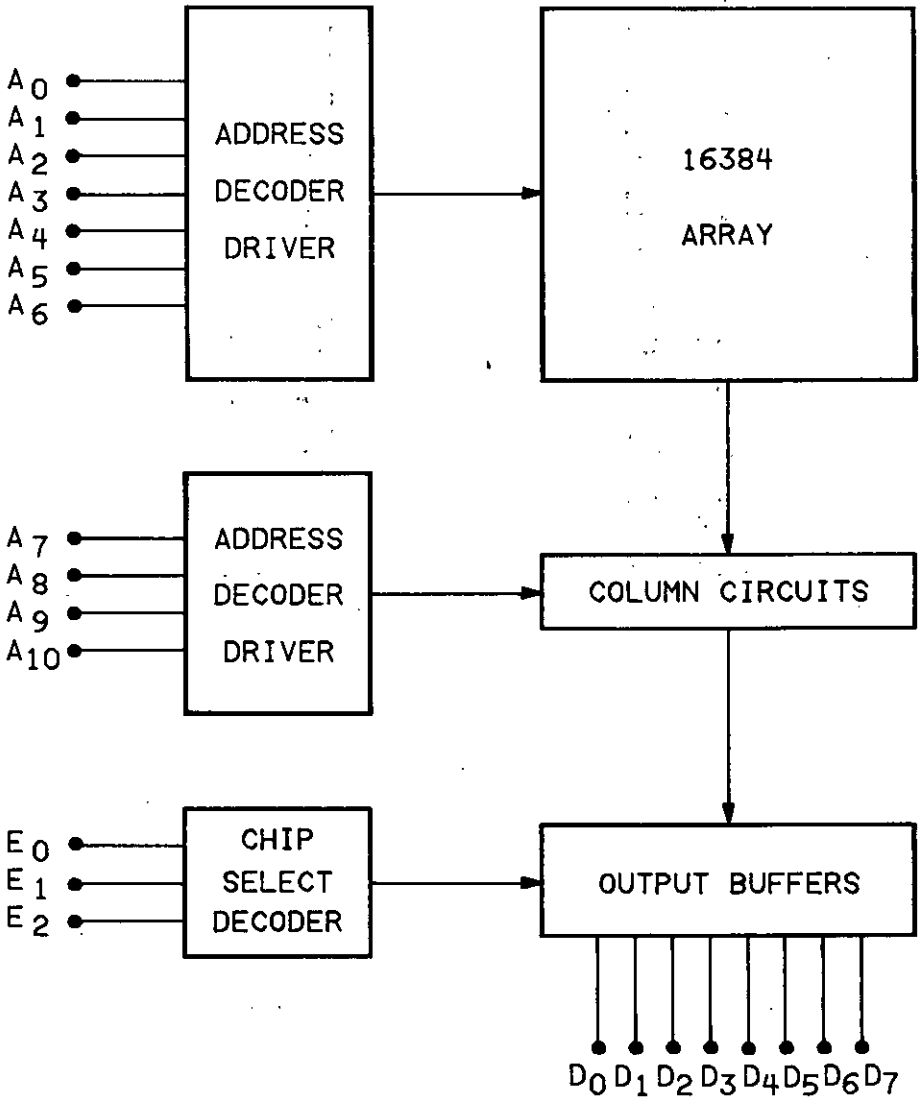
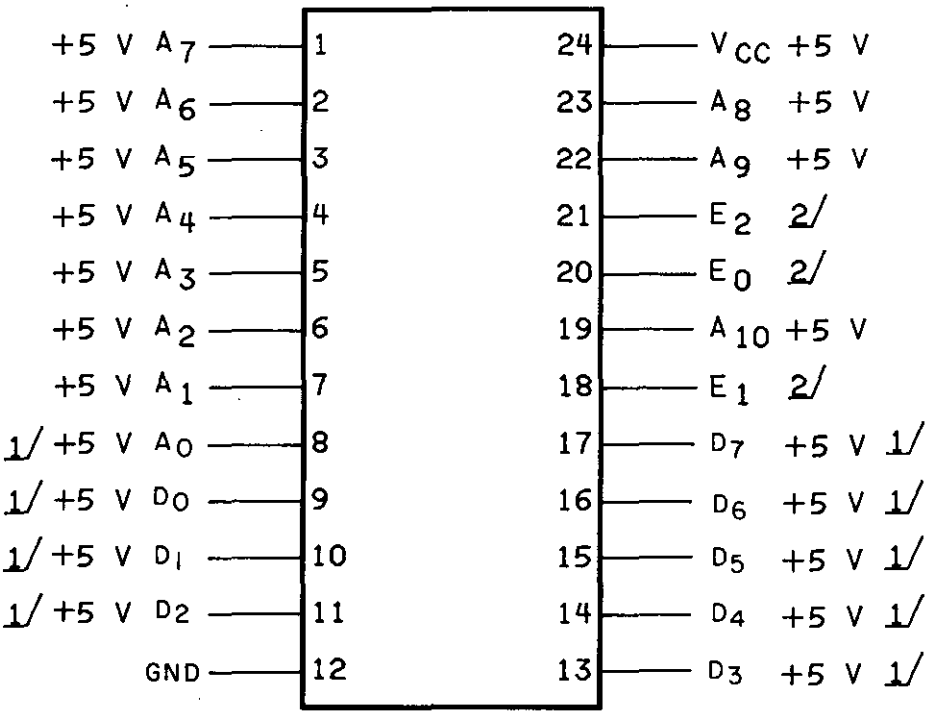


FIGURE 2. Block diagram.

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NOTES:
1/ At the manufacturer's option, a 1 kΩ resistor to V_{CC} may be used on the output terminals.
2/ Enable inputs should be biased in the disable mode.

FIGURE 3. Static burn-in test circuit.

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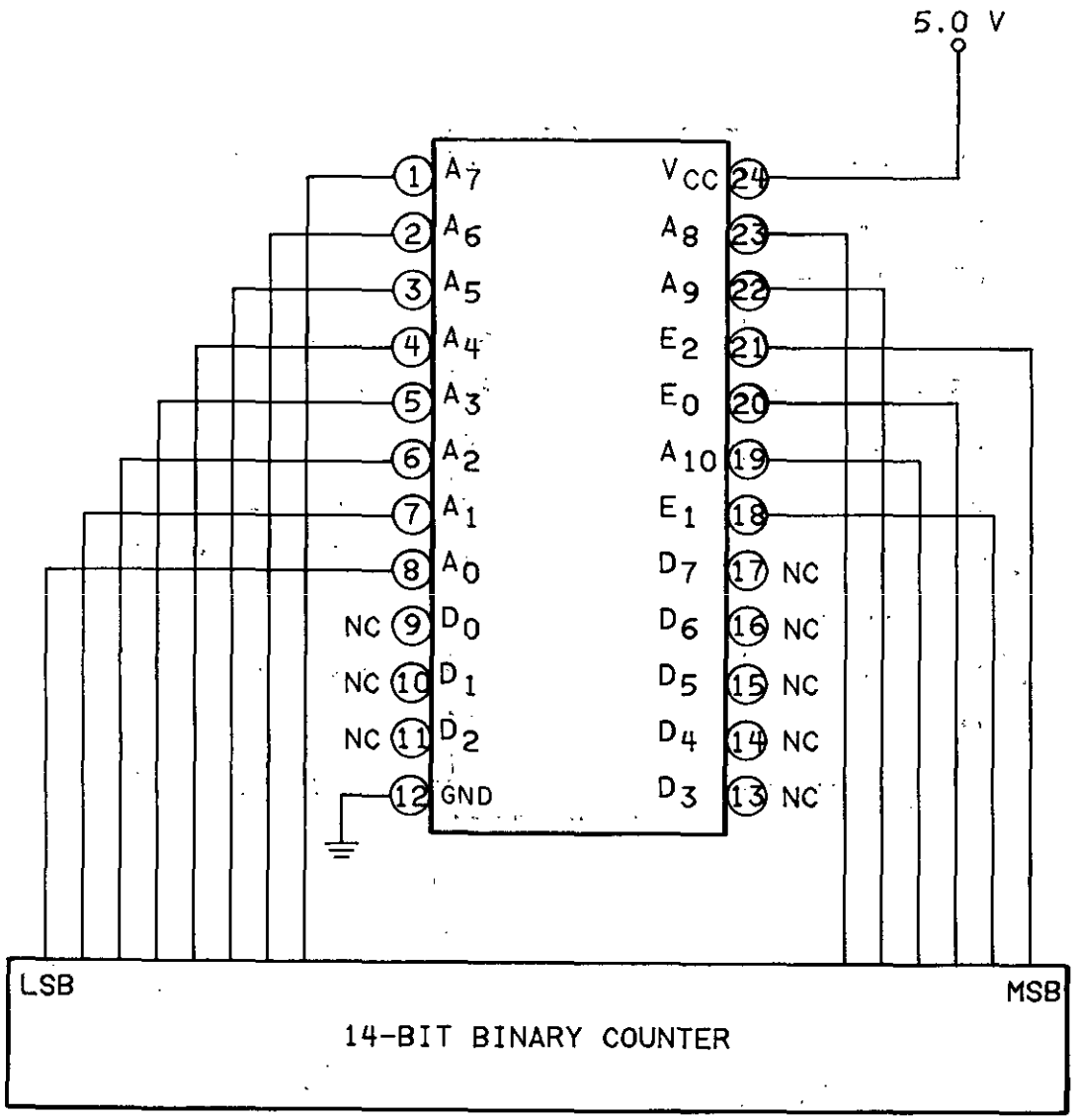


FIGURE 4. Dynamic burn-in and life test circuit.

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TABLE III. Group A inspection or device type 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Case J	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits		Unit
			Test no.	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	D_0	D_1	D_2	GND	D_3	D_4	D_5	D_6	D_7	E_1	A_{10}	E_0	E_2	A_9	A_8	V_{CC}		Min	Max	
T_C = +25°C 1	I_{IL} 1/	3009	1	GND	GND	GND	GND	GND	GND	GND	GND				GND							GND			GND	GND	5.5 V	A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0 A_{10} A_9 A_8		2.5	μA
	I_{IH} 1/	3010	12	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V											5.5 V			5.5 V	5.5 V		A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0 A_{10} A_9 A_8			μA
	I_{OLZ}	3007	23									0.4 V	0.4 V	0.4 V		0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	2/		2/	2/				D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7		10	
	I_{OHZ}		31									2.4 V	2.4 V	2.4 V		2.4 V	2.4 V	2.4 V	2.4 V	2.4 V								D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7			
	I_{CC}	3005	39																									V_{CC}		130	mA
	V_{OH} 3/	3006	40									-100 μA	-100 μA	-100 μA		-100 μA	-100 μA	-100 μA	-100 μA	-100 μA								D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7	2.4		V
	V_{OL} 4/		48									1.6 mA	1.6 mA	1.6 mA		1.6 mA	1.6 mA	1.6 mA	1.6 mA	1.6 mA								D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7		0.4	

See footnotes at end of table.

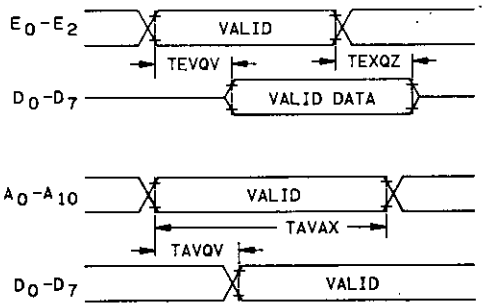
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TABLE III. Group A inspection or device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; low ≤ 0.7 V; or open)

Subgroup	Symbol	MIL-STD-883 method	Case J	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	D_0	D_1	D_2	GND	D_3	D_4	D_5	D_6	D_7	E_1	A_{10}	E_0	E_2	A_9	A_8	V_{CC}				
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C.																														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C.																														
4 T_C = +25°C	C_I 5/	3012	56	5/	5/	5/	5/	5/	5/	5/	5/				GND							5/			5/	5/	GND	A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0 A_{10} A_9 A_8		7.5	pF
		"	57												"												"			"
		"	58												"												"			"
		"	59												"												"			"
		"	60												"												"			"
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	C_O 5/	"	67									5/	5/	5/	"	5/	5/	5/	5/	5/							"	D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7		12.5	pF
		"	68												"												"			"
		"	69												"												"			"
		"	70												"												"			"
		"	71												"												"			"
		"	72												"												"			"
		"	73												"												"			"
		"	74												"												"			"
7 T_C = +25°C	Functional tests 6/7/8/9/	3014	75												"												7/	D_0 thru D_7			
8	Same tests and terminal conditions as for subgroup 7, except T_C = +125°C and -55°C.																														
9 T_C = +25°C 6/7/8/9/	TAVAX	3003	76												GND												7/	All outputs	450		ns
	TAVQV	"	77												"												"			450	"
	TEVQV	"	78												"												"			200	"
	TEXQZ	"	79												"												"		10	150	"
10	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = +125°C.																														
11	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = -55°C.																														

1/ Leakage current (I_{IH}, I_{IL}) subgroup 1, shall not be required for 100% screening.
2/ Outputs driven to high Z state using E_0, E_1, and E_2 (logic level is customer defined during mask-program definition).
3/ Address shall be selected from user mask-program for high output of the tested pin.
4/ Address shall be selected from user mask-program for low output of the tested pin.
5/ The capacitance measurements shall be made between the indicated terminal and the grounded terminals at a frequency of 1 MHz. The dc bias of the measuring instrument shall be less than ±0.1 V. The ac signal amplitude shall be less than 50 mV rms.
6/ Input and output logic levels for functional tests are derived from a custom ROM test pattern unique to each ROM variant. The test pattern shall include a truth table for the 2048 byte data array, and a truth table for the 8 combinations of E_0, E_1, and E_2.
7/ For V_{CC} = 5.5 V or 4.5 V, V_{IL} = 0.8 V, V_{IH} = 2.0 V, V_{OL} <0.4 V, V_{OH} >2.4 V. Tests for subgroups 7, 8, 9, 10, and 11 shall be accomplished with V_{CC} at 4.5 V and 5.5 V.
8/ GALPAT pattern description.
1. Word 0 is read
2. Word 1 is read
3. Word 0 is read
4. Word 2 is read
5. Word 0 is read
6. The reading procedure continues back and forth between the word 0 and the next higher numbered word until word 2047 is reached, then increments to the next word and reads back and forth as in steps 1 through 6 until all words are tested.

9/ Timing diagram:



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4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4). Generic test data (see 6.6) may be used to satisfy the requirements for groups C and D inspections. Quality conformance inspection shall be completed on the specific devices covered by this specification before they are shipped.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroup 4 (C_i and C_o measurements) shall be performed only after process or design changes which may affect design capacitance.

c. Subgroups 5 and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein. A special subgroup shall be added using a lot tolerance percent defective (LTPD) of 15 for class B devices. This subgroup shall consist of a high voltage test of the input protection circuits, V_{ZAP} (see 4.5.3).

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. Subgroups 3, 4, and 5 shall be added to the group C inspection requirements for class B devices, and shall consist of the tests, conditions, and limits specified for subgroups 9, 10, and 11 of group A. These additional tests shall consist of performing the specified tests using the GALPAT pattern as specified in table III.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

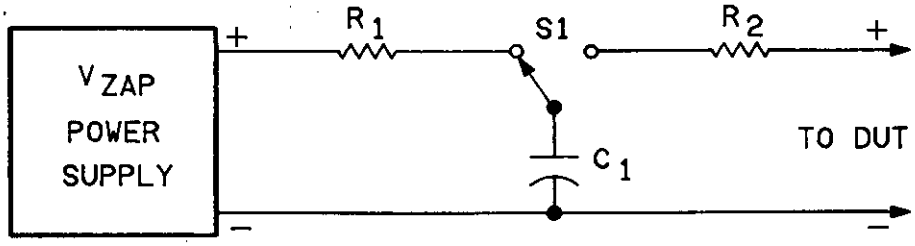
4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 Life test cooldown procedure. When devices are measured at +25°C following application of the steady-state life or burn-in test condition, they shall be cooled to 35°C or less, prior to removal of bias.

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4.5.3 High voltage (V_{ZAP}) test of input protection circuits. All input terminals (up to a maximum of four) of the device under test (DUT) shall be subjected to a voltage pulse from a 100 pF source charged to 150 volts. This destructive test shall be conducted in accordance with the following, using the test circuit shown on figure 5.



R_1 = 10 MΩ ≤ R1 ≤ 50 MΩ.
R_2 = 1.5 kΩ.
C_1 = 100 pF.
V_{ZAP} = 150 V charge on C1.
S_1 = Hg-wetted "bounceless" relay.

FIGURE 5. High voltage (V_{ZAP}) test circuits.

a. Measure I_{IH} and I_{IL} at inputs selected as stated above, at 25°C and V_{CC} = 5.0 volts. These measurements shall be made in accordance with table III herein. The test limits for a single terminal measurement of I_{IH} and I_{IL} shall be 10 microamperes maximum.

b. V_{ZAP} is applied to the DUT in the following modes (see table V) by charging C1 to V_{ZAP} with S1 in position 1 and then switching to position 2.

TABLE V. Modes for high-voltage test.

Mode	+ Terminal	- Terminal
1	V_{CC}	Input
2	Input	GND
3	Input	Associated output

c. Within 24 hours, repeat the I_{IH} and I_{IL} measurements on the same terminals as performed above. If a DUT exhibits leakage currents in excess of the specified limits after the V_{ZAP} test, it shall be classified as a failure.

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4.6 Data reporting. When specified in the contract or purchase order, a copy of the following data, as applicable, shall be supplied:

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and steady-state life tests.

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in.

e. Final electrical parameters data (see 4.2d).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be as specified in MIL-M-38510. The devices covered by this specification require electrostatic protection.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity if applicable.

e. Requirements for packaging and packing.

f. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

A_0 thru A_{10}	Address inputs.
C_i	Total input capacitance.
C_o	Total output capacitance.
D_0 thru D_7	Data outputs.
GND	Electrical ground (common terminal).
I_{IH}, I_{IL}	Input leakage current for high voltage and low voltage, respectively.
I_{OHZ}, I_{OLZ}	Output high impedance leakage current for high voltage and low voltage, respectively.

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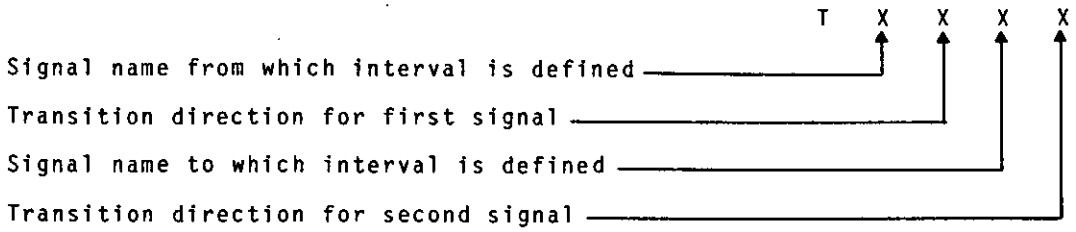
TAVAX - - - - - - - - - - - - - - - - Read cycle time.

TAVQV - - - - - - - - - - - - - - - - Address access time.

TEVQV - - - - - - - - - - - - - - - - Enable access time.

TEXQZ - - - - - - - - - - - - - - - - Output disable time.

6.4.1 Timing parameter abbreviations. All timing abbreviations use uppercase characters with no subscripts. The initial character is always "T" and is followed by four descriptors. These characters specify two signal points arranged in a "from-to" sequence that define a timing interval. The two descriptors for each signal point specify the signal name and the signal transitions. Thus the format is:



a. Signal definitions:
   A = Address
   D = Data in
   E = Chip enable
   G = Output enable
   Q = Data out
   S = Chip select
   W = Write enable

b. Transition definitions:
   H = Transition to high
   L = Transition to low
   V = Transition to valid
   X = Transition to invalid or don't care
   Z = Transition to off (high impedance)

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer lead lengths and lead forming shall not affect the part number.

6.6 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes no more than one year old and on a die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the same plant as the die represented. Group D generic data shall be on date codes no more than one year old and on the same package type (see MIL-M-38510) and from the same plant as the package represented. The vendor is required to retain the generic data for a period of not less than 36 months from the date of shipment.

6.7 Handling. NMOS microcircuits should be handled with certain precautions to avoid damage due to the accumulation of static charge. These NMOS devices are fabricated with a silicon gate technology, including input protection, which reduces the susceptibility to damage. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

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d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in areas where NMOS devices are handled.

f. Maintain relative humidity above 50 percent, if practical.

6.8 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	S6831B, 68A316E, 9218, 3516E, 2316E, MK34000, 52116, 2616

6.9 Ordering guidance. Since the qualification and certification requirements have been removed from the specification, orders may be placed immediately.

6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC,

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/403A	MICROCIRCUITS, DIGITAL, NMOS, 2048-WORD x 8-BIT STATIC READ ONLY

3a. NAME OF SUBMITTING ORGANIZATION MEMORY (ROM), MONOLITHIC	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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PREVIOUS EDITION IS OBSOLETE.

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